MU_3SL Shared Library AN1

Offline Calibration

preliminary

SCOPE OF THE DOCUMENT

This document is intended to support software developers during the iC-MU Series design-in process. It describes the steps recommended by iC-Haus for a 2-track offline calibration (analog and Nonius) and refers to the example **calibration_offline_data_file.c**

Note:
In this document only functions are used that do not require direct communication to the iC-MU Series chip via an iC-Haus PC USB adapter (e.g. MB5U).

This document is only applicable for iC-MU shared library vision 3.x.x or later.

OFFLINE CALIBRATION: SETUP EXAMPLE

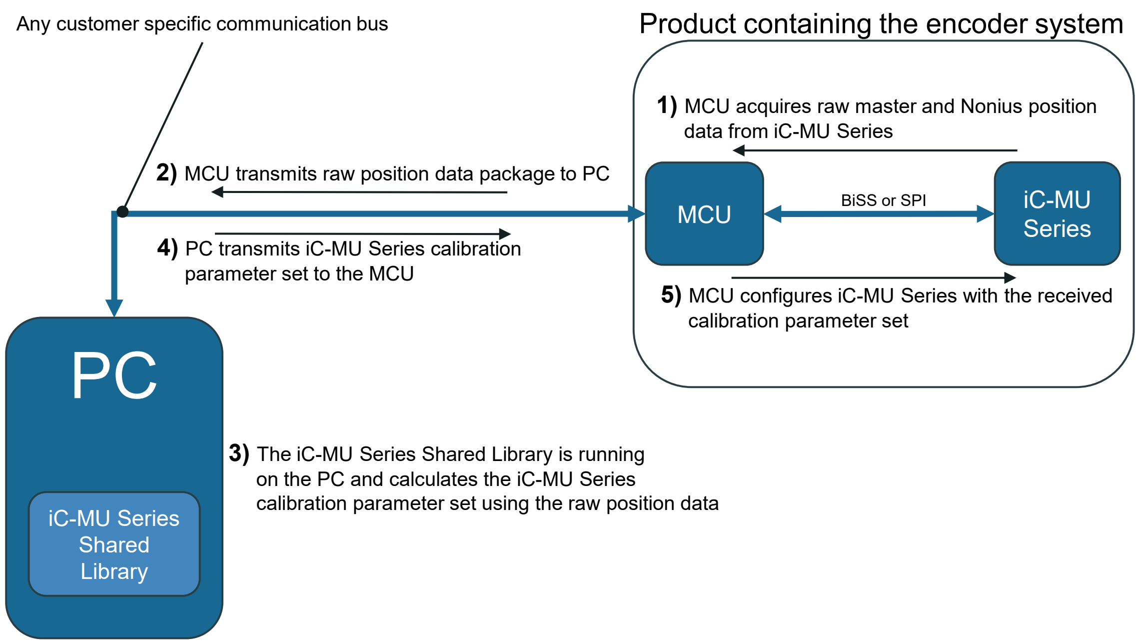
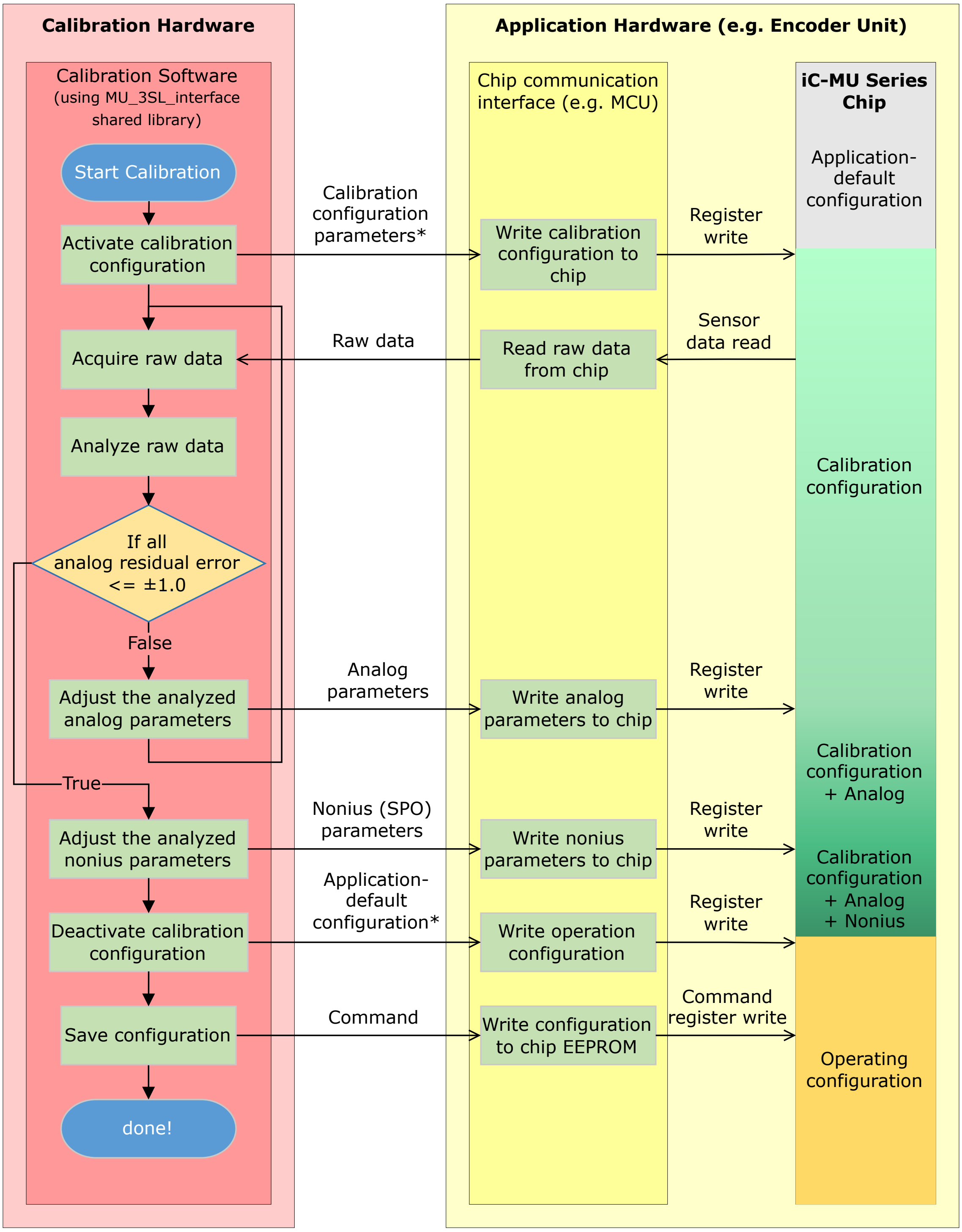


Figure 1: Offline Calibration Hardware Data Flow Diagram

OFFLINE CALIBRATION SOFTWARE FLOW CHART



* Set consisting of serial output parameters and the ENAC parameter.

Figure 2: Illustration of the offline calibration procedure

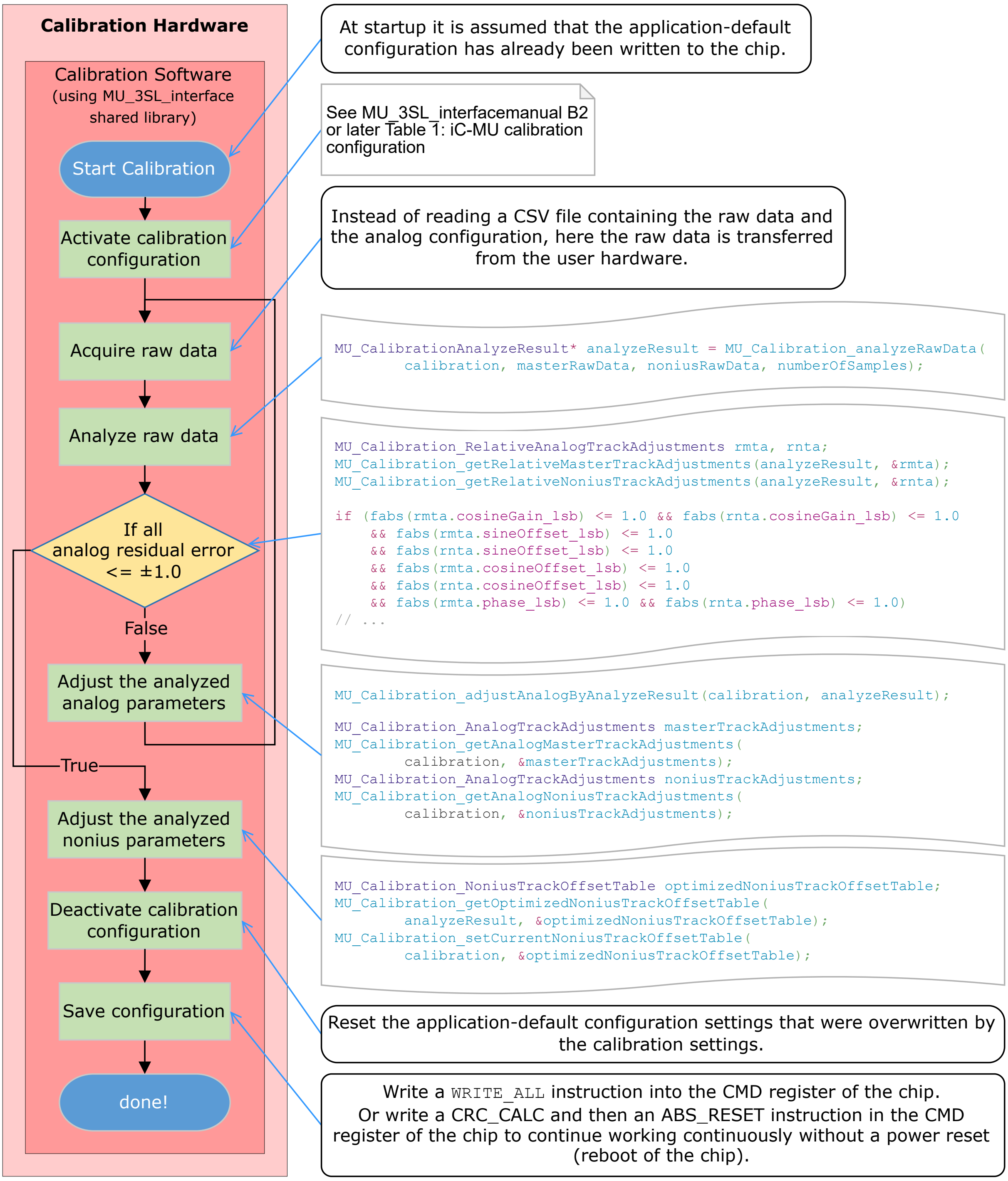


Figure 3: Flow Chart Notes

DESCRIPTION OF CONFIGURATION STATES

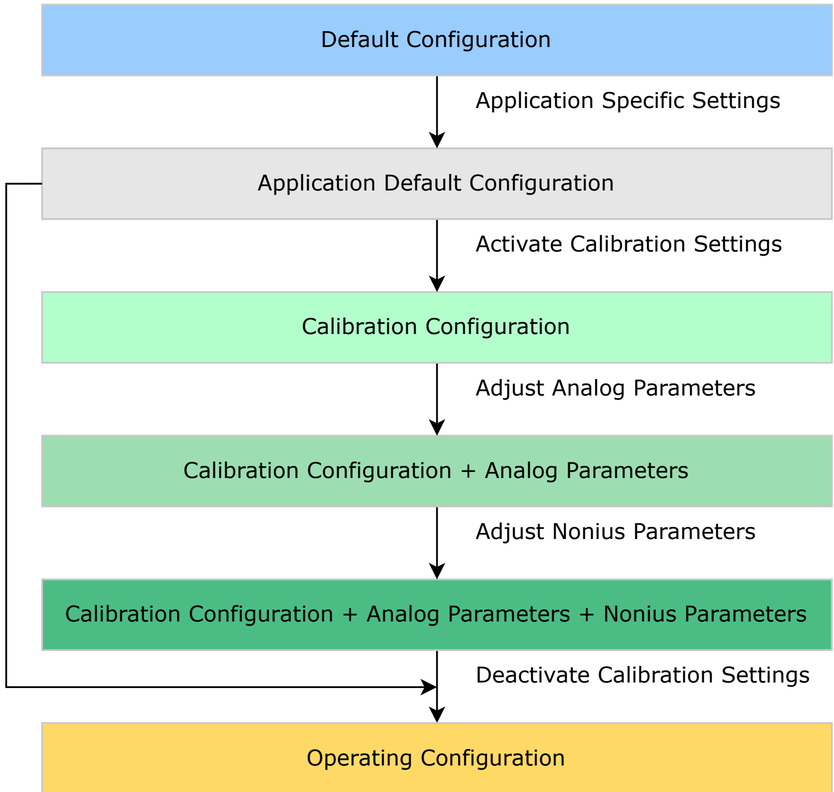


Figure 4: Running through the iC-MU Series configuration states during calibration

Default Configuration
This is the configuration that iC-MU Series assumes after power-on. In order to communicate with iC-MU Series it needs to be ensured, that MODEA is set to the required interface (MODEA = 0x02: BiSS or MODEA = 0x00: SPI) in the default configuration. All other parameter settings are irrelevant for the offline calibration, as they will be overwritten by the application default configuration.

Application Default Configuration
This configuration is defined by the user of iC-MU Series and tailored towards the specific application. It does not yet contain the analog and Nonius calibration parameters. Those parameter are to be determined and transferred to the iC-MU Series chip during the calibration process.

Calibration Configuration
This configuration is used to record the position data in a defined format that is required by the shared library calibration functions. For this purpose, the application default configuration is modified as follows:

Chip Parameter	Value
ENAC	0x01
OUT_LSB	0x00
OUT_MSB	0x0E
OUT_ZERO	If connected via SPI then 0x04, else 0x00
MODE_ST	0x02
TEST	0x00
MPC	If MPC == 0x0C then 0x0B, else not changed

Table 1: Calibration Settings

Calibration Configuration + Analog Parameters
This is the calibration configuration amended by the analog parameters (see Table 1) that are determined by the shared library calibration algorithm.

Calibration Configuration + Analog Parameters + Nonius Parameters
This is the calibration configuration with the analog parameters from the previous step amended by the Nonius parameters (see Table 2) that are determined by the shared library calibration algorithm.

Operating Configuration
After the calibration is finalized the calibration configuration is deactivated with the respective shared library function (see Figure 3). To this end the application default configuration that was overwritten by the calibration configuration is restored. The analog and Nonius parameters determined in the previous steps are retained. The operating configuration is the final configuration for the series application of the iC-MU Series based encoder system.

Analog Parameters								
Addr. SER	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
0x01		GX_M(6:0)						
0x02		VOSS_M(6:0)						
0x03		VOSC_M(6:0)						
0x04	PHR_M	PH_M(6:0)						
0x07		GX_N(6:0)						
0x08		VOSS_N(6:0)						
0x09		VOSC_N(6:0)						
0x0A	PHR_N	PH_N(6:0)						

Table 2: iC-MU Series analog parameters adjusted by the shared library calibration algorithm

Nonius Parameters								
Addr. SER	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
0x52	SPO_0(3:0)				SPO_BASE(3:0)			
0x53	SPO_2(3:0)				SPO_1(3:0)			
0x54	SPO_4(3:0)				SPO_3(3:0)			
0x55	SPO_6(3:0)				SPO_5(3:0)			
0x56	SPO_8(3:0)				SPO_7(3:0)			
0x57	SPO_10(3:0)				SPO_9(3:0)			
0x58	SPO_12(3:0)				SPO_11(3:0)			
0x59	SPO_14(3:0)				SPO_13(3:0)			

Table 3: iC-MU Series Nonius parameters adjusted by the shared library calibration algorithm

REVISION HISTORY

Rev.	Notes	Pages affected	DLL rev.
A1	Initial version	All pages	3.x.x